CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

PART – 1

INTRODUCTION TO CHEMICAL REACTIONS

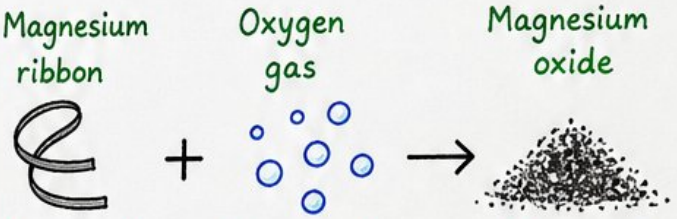
★ 1. WHAT IS A CHEMICAL REACTION?

A chemical reaction is a process in which one or more substances, called reactants, are converted into one or more new substances, called products, with different properties.

Example : Burning of magnesium

$$2Mg + O_2 \longrightarrow 2MgO$$

(Reactants) (Product)

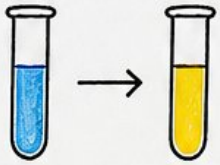


In this reaction, magnesium (Mg) reacts with oxygen (O_2) to form magnesium oxide (MgO).

★ 2. CHARACTERISTICS OF CHEMICAL REACTIONS

We can identify a chemical reaction by one or more of the following characteristics :

① Change in colour



For example, blue copper sulphate solution turns green when iron is added.

② Evolution of a gas



For example, bubbles of carbon dioxide gas are produced when marble chips react with dilute HCl.

③ Change in temperature



For example, temperature rises when quicklime (CaO) is mixed with water.

④ Formation of a precipitate



For example, white precipitate of silver chloride (AgCl) is formed when silver nitrate reacts with sodium chloride.

⑤ Change in state



For example, ice changes into water on melting (endothermic process).

NOTE :
- Not every physical change is a chemical reaction.
- In a chemical reaction, new substances with new properties are formed.

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★ 3. EXAMPLES OF CHEMICAL REACTIONS IN DAILY LIFE

- Burning of candle → Wax reacts with oxygen to give carbon dioxide and water.
- Rusting of iron → Iron reacts with oxygen and moisture to form rust (iron oxide).
- Digestion of food → Complex food breaks down into simpler substances in our body.
- Souring of milk → Milk changes into curd due to the action of bacteria.
- Photosynthesis → Green plants use CO_2 and water to make food in presence of sunlight.

★ CHEMICAL REACTIONS ARE THE BASIS OF MANY PROCESSES AROUND US. ★

CLASS 10 SCIENCE
CHEMICAL REACTIONS AND EQUATIONS

PART – 2 (CHEMICAL EQUATIONS)

★ 1. WHAT IS A CHEMICAL EQUATION ?

A chemical equation is a symbolic representation of a chemical reaction in the form of symbols and chemical formulae of the reactants and products.

Example : Formation of water

Hydrogen + Oxygen ⟶ Water

$H_2 + O_2 \longrightarrow H_2O$

Reactants | Product

★ 2. TYPES OF CHEMICAL EQUATIONS

(i) WORD EQUATION

A chemical equation in which the names of reactants and products are written.

Example :

Hydrogen + Oxygen ⟶ Water

(ii) SKELETAL EQUATION

A chemical equation in which the formulae of reactants and products are written without balancing.

Example :

$H_2 + O_2 \longrightarrow H_2O$

(iii) BALANCED CHEMICAL EQUATION

A chemical equation in which the number of atoms of each element is equal on both sides of the equation.

Example :

$2H_2 + O_2 \longrightarrow 2H_2O$

★ 3. IMPORTANCE OF CHEMICAL EQUATIONS

- They tell us what are the reactants and products.
- They show the physical states of substances.
- They give information about the proportion of substances taking part in a reaction.
- They help us to calculate the amount of reactants needed or products formed.
- They help in understanding and predicting chemical reactions.

PHYSICAL STATES

(s) = Solid
(ℓ) = Liquid
(g) = Gas
(aq) = Aqueous (dissolved in water)

Example :

$H_{2(g)} + O_{2(g)} \longrightarrow H_2O_{(\ell)}$

★ 4. EXAMPLES OF CHEMICAL EQUATIONS

S. No.	Reaction	Word Equation	Skeletal Equation	Balanced Chemical Equation
1.	Burning of Magnesium	Magnesium + Oxygen ⟶ Magnesium oxide	$Mg + O_2 \longrightarrow MgO$	$2Mg + O_2 \longrightarrow 2MgO$
2.	Formation of Water	Hydrogen + Oxygen ⟶ Water	$H_2 + O_2 \longrightarrow H_2O$	$2H_2 + O_2 \longrightarrow 2H_2O$
3.	Reaction of Zinc with Sulphuric Acid	Zinc + Sulphuric acid ⟶ Zinc sulphate + Hydrogen	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$
4.	Thermal Decomposition of Calcium Carbonate	Calcium carbonate ⟶ Calcium oxide + Carbon dioxide	$CaCO_3 \longrightarrow CaO + CO_2$	$CaCO_3 \longrightarrow CaO + CO_2$

NOTE : Always balance a chemical equation to obey the law of conservation of mass (Matter is neither created nor destroyed).



CLASS 10 SCIENCE
CHEMICAL REACTIONS AND EQUATIONS



PART – 3
BALANCED CHEMICAL EQUATIONS

★ 1. WHAT IS A BALANCED CHEMICAL EQUATION ?

A balanced chemical equation is one in which the number of atoms of each element is equal on both sides of the equation.

Example :

Unbalanced : $H_2 + O_2 \longrightarrow H_2O$

Balanced : $2H_2 + O_2 \longrightarrow 2H_2O$

Why balance ?

- To obey the law of conservation of mass.
- To know the exact proportion of reactants and products.
- Helpful in calculations involving chemical reactions.

LAW OF CONSERVATION OF MASS

Mass can neither be created nor destroyed in a chemical reaction.



★ 2. STEPS TO BALANCE A CHEMICAL EQUATION

① Write the skeleton (unbalanced) equation.
② Count the number of atoms of each element on both sides.
③ Balance the atoms one by one. (Start with the element which appears in least number of compounds.)
④ First balance metals, then non-metals.
⑤ Then balance hydrogen atoms.
⑥ Finally balance oxygen atoms.
⑦ Check the number of atoms again. Both sides must be equal.
⑧ Do not change the formula of the compounds, only put coefficients.

IMPORTANT TIPS

★ Never change the subscripts.
★ Only change the coefficients.
★ Use the smallest whole number coefficients.
★ If a polyatomic ion appears unchanged on both sides, treat it as a unit.

★ 3. BALANCING OF CHEMICAL EQUATIONS – EXAMPLES

① Hydrogen + Oxygen → Water

Skeleton equation : $H_2 + O_2 \longrightarrow H_2O$
Step 1 : H → 2 on LHS, 2 on RHS ✓
Step 2 : O → 2 on LHS, 1 on RHS ✗
Step 3 : Put 2 before H_2O → $H_2 + O_2 \longrightarrow 2H_2O$
Step 4 : Now O → 2 and ✓
Balanced equation : $2H_2 + O_2 \longrightarrow 2H_2O$

② Magnesium + Oxygen → Magnesium oxide

Skeleton equation : $Mg + O_2 \longrightarrow MgO$
Step 1 : O → 2 on LHS, 1 on RHS ✗
Step 2 : Put 2 before MgO → $Mg + O_2 \longrightarrow 2MgO$
Step 3 : Mg → 1 on LHS, 2 on RHS ✗
Step 4 : Put 2 before Mg → $2Mg + O_2 \longrightarrow 2MgO$
Balanced equation : $2Mg + O_2 \longrightarrow 2MgO$

③ Iron + Sulphur → Iron(II) sulphide

Skeleton equation : $Fe + S \longrightarrow FeS$
Step 1 : Fe → 1, S → 1 on both sides ✓
Balanced equation : $Fe + S \longrightarrow FeS$

(Already balanced)

④ Aluminium + Oxygen → Aluminium oxide

Skeleton equation : $Al + O_2 \longrightarrow Al_2O_3$
Step 1 : O → 2 on LHS, 3 on RHS ✗
Step 2 : LCM of 2 and 3 = 6
Put 3 before O_2 and 2 before Al_2O_3 →
Step 3 : Now O → 6 and 6 ✓
Step 4 : Al → 1 on LHS, 4 on RHS ✗
Step 5 : Put 4 before Al → $4Al + 3O_2 \longrightarrow 2Al_2O_3$
Balanced equation : $4Al + 3O_2 \longrightarrow 2Al_2O_3$

★ 4. PRACTICE YOURSELF

Balance the following equations :

(i) $N_2 + H_2 \longrightarrow NH_3$
(ii) $KClO_3 \longrightarrow KCl + O_2$
(iii) $CaCO_3 \longrightarrow CaO + CO_2$
(iv) $NaOH + HCl \longrightarrow NaCl + H_2O$
(v) $Zn + HCl \longrightarrow ZnCl_2 + H_2$
(vi) $CH_4 + O_2 \longrightarrow CO_2 + H_2O$



★ A balanced equation is the key to understanding and calculating a chemical reaction. ★



CLASS 10 SCIENCE
CHEMICAL REACTIONS AND EQUATIONS



PART – 4

TYPES OF CHEMICAL REACTIONS

Chemical reactions are of different types depending on the number and nature of reactants and products formed.

1. COMBINATION REACTION

Two or more substances combine to form a single product.

General form : $A + B \longrightarrow AB$

Examples :

(i) $2Mg + O_2 \longrightarrow 2MgO$

Magnesium + Oxygen → Magnesium oxide

(ii) $CaO + H_2O \longrightarrow Ca(OH)_2$

Calcium oxide + Water → Calcium hydroxide

2. DECOMPOSITION REACTION

A single compound breaks down into two or more simpler substances.

General form : $AB \longrightarrow A + B$

Examples :

(i) $CaCO_3 \xrightarrow{\Delta\ (Heat)} CaO + CO_2$

Calcium carbonate → Calcium oxide + Carbon dioxide

(ii) $2H_2O \xrightarrow{Electricity} 2H_2 + O_2$

Water → Hydrogen + Oxygen

(iii) $2AgCl \xrightarrow{Sunlight} 2Ag + Cl_2$

Silver chloride → Silver + Chlorine

3. DISPLACEMENT REACTION

A more reactive element displaces a less reactive element from its compound.

General form : $A + BC \longrightarrow AC + B$

Examples :

(i) $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$

Zinc + Copper sulphate → Zinc sulphate + Copper sulphate

(ii) $Fe + 2HCl \longrightarrow FeCl_2 + H_2$

Iron + Hydrochloric acid → Iron(II) chloride + Hydrogen

4. DOUBLE DISPLACEMENT REACTION

The positive ions and negative ions of two compounds exchange places to form two new compounds.

General form : $AB + CD \longrightarrow AD + CB$

Examples :

(i) $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 \downarrow + 2NaCl$

Sodium sulphate + Barium chloride → Barium sulphate ↓ (White precipitate) + Sodium chloride

(ii) $AgNO_3 + NaCl \longrightarrow AgCl \downarrow + NaNO_3$

Silver nitrate + Sodium chloride → Silver chloride ↓ (White precipitate) + Sodium nitrate

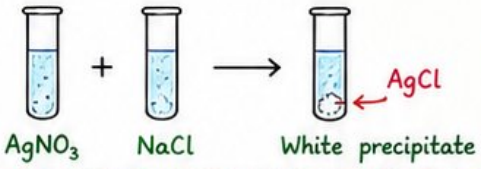
5. PRECIPITATION REACTION

A type of double displacement reaction in which an insoluble solid (precipitate) is formed.

Example :

$AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$

Silver nitrate + Sodium chloride → Silver chloride + Sodium nitrate



Insoluble precipitate is indicated by ↓

IMPORTANT NOTE

- ★ All precipitation reactions are double displacement reactions
- ★ but all double displacement reactions are not precipitation reactions.

QUICK RECAP

1. Combination — $A + B \longrightarrow AB$
2. Decomposition — $AB \longrightarrow A + B$
3. Displacement — $A + BC \longrightarrow AC + B$
4. Double Displacement — $AB + CD \longrightarrow AD + CB$
5. Precipitation (a type of double displacement)

REMEMBER

Chemical reactions are the basis of many natural phenomena and industrial processes.

★ Understand each type with examples and practise writing balanced equations. ★

CLASS 10 SCIENCE
CHEMICAL REACTIONS AND EQUATIONS

PART – 5

EXOTHERMIC AND ENDOTHERMIC REACTIONS

Chemical reactions are accompanied by absorption or evolution of heat energy. On this basis, reactions are of two types :

1. EXOTHERMIC REACTIONS



- ★ Reactions in which heat energy is released to the surroundings.
- ★ The temperature of the surroundings increases.
- ★ These reactions are generally fast.

Examples :

(i) Burning of natural gas (methane)

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O + \text{Heat}$$

(ii) Neutralisation reaction (acid + base)

$$HCl + NaOH \longrightarrow NaCl + H_2O + \text{Heat}$$

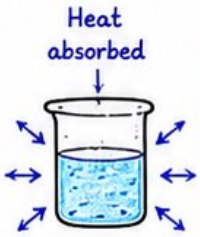
(iii) Combustion of fuels

$$C + O_2 \longrightarrow CO_2 + \text{Heat}$$

(iv) Reaction of quicklime with water

$$CaO + H_2O \longrightarrow Ca(OH)_2 + \text{Heat}$$

2. ENDOTHERMIC REACTIONS



- ★ Reactions in which heat energy is absorbed from the surroundings.
- ★ The temperature of the surroundings decreases.
- ★ These reactions are generally slow.

Examples :

(i) Thermal decomposition of calcium carbonate

$$CaCO_3 \xrightarrow{\Delta\ (\text{Heat})} CaO + CO_2$$

(ii) Photosynthesis in green plants

$$6CO_2 + 6H_2O \xrightarrow{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$$

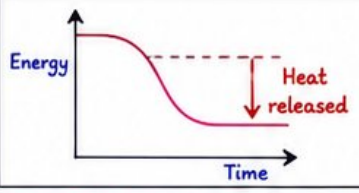
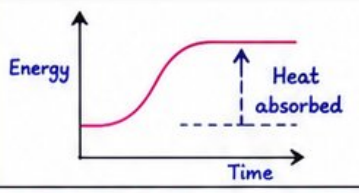
(iii) Dissolution of ammonium chloride in water

$$NH_4Cl + \text{Heat} \longrightarrow NH_4Cl\ (\text{in solution})$$

(iv) Electrolysis of water

$$2H_2O \xrightarrow{\text{Electricity}} 2H_2 + O_2$$

DIFFERENCES BETWEEN EXOTHERMIC AND ENDOTHERMIC REACTIONS

Basis	Exothermic Reactions	Endothermic Reactions
1. Heat energy	Heat is released.	Heat is absorbed.
2. Effect on surroundings	Surroundings get heat.	Surroundings lose heat.
3. Temperature	Temperature increases.	Temperature decreases.
4. Nature	Generally fast.	Generally slow.
5. Examples	Combustion, neutralisation, respiration, reaction of CaO with water.	Thermal decomposition of $CaCO_3$, photosynthesis, dissolution of NH_4Cl.
6. Graph		

REAL LIFE EXAMPLES

Exothermic :

When we burn LPG in the stove, heat is released and food gets cooked.



Endothermic :

When we apply a cold pack (containing ammonium nitrate and water), it absorbs heat from the body and gives a cooling effect.



IMPORTANT POINTS

- ✔ No reaction can take place without absorption or evolution of energy.
- ✔ Exothermic reactions release heat, so they are used in heaters, cooking, etc.
- ✔ Endothermic reactions absorb heat, so they are used in cold packs, refrigerants, etc.
- ✔ Energy is always conserved in a chemical reaction.

KEYWORDS

Exothermic → Heat released

Endothermic → Heat absorbed

ENERGY CHANGES ARE AN IMPORTANT PART OF EVERY CHEMICAL REACTION.

CLASS 10 SCIENCE
CHEMICAL REACTIONS AND EQUATIONS

PART – 6

OXIDATION AND REDUCTION (REDOX REACTIONS)

1. OXIDATION

Oxidation is the process in which a substance :

- ★ Adds oxygen
  OR
- ★ Loses hydrogen
  OR
- ★ Loses electrons

Remember : **OIL** — Oxidation Is Loss

Examples :

(i) Magnesium burns in air.

$$2Mg + O_2 \longrightarrow 2MgO$$

(Magnesium adds oxygen → oxidised)

(ii) Copper(II) oxide is reduced by hydrogen.

$$CuO + H_2 \longrightarrow Cu + H_2O$$

(Hydrogen is lost by CuO → oxidised)

2. REDUCTION

Reduction is the process in which a substance :

- ★ Removes oxygen
  OR
- ★ Gains hydrogen
  OR
- ★ Gains electrons

Remember : **RIG** — Reduction Is Gain

Examples :

(i) Copper(II) oxide is reduced by hydrogen.

$$CuO + H_2 \longrightarrow Cu + H_2O$$

(CuO removes oxygen → reduced)

(iii) Zinc oxide is reduced by carbon.

$$ZnO + C \longrightarrow Zn + CO$$

(ZnO removes oxygen → reduced)

3. REDOX REACTION

A reaction in which oxidation and reduction occur simultaneously is called a redox (reduction–oxidation) reaction.

EXAMPLE :

$$\underbrace{CuO}_{\text{Reduced}} + H_2 \longrightarrow Cu + \underbrace{H_2O}_{\text{Oxidised}}$$

(Oxidised: H_2 → H_2O; Reduced: CuO → Cu)

CuO → Loses oxygen → Reduced

H_2 → Gains oxygen → Oxidised

4. OXIDATION NUMBERS (OXIDATION STATE)

The charge that an atom would have if all bonds with atoms of different elements are 100% ionic.

Rules :

1. The oxidation number of an element in free state = 0 (e.g., Na, O_2, H_2, Cl_2, etc.)
2. The oxidation number of a monatomic ion = its charge (e.g., $Na^+ = +1$, $Cl^- = -1$)
3. The sum of oxidation numbers of all atoms in a neutral compound = 0
4. The sum of oxidation numbers of all atoms in a polyatomic ion = charge on the ion
5. Group 1 metals = +1, Group 2 metals = +2
6. Oxygen = −2 (except in peroxides −1 and in OF_2 +2)
7. Hydrogen = +1 (except in metal hydrides −1)

5. SIMULTANEOUS OXIDATION AND REDUCTION

In a redox reaction :

- One substance is oxidised and another is reduced.
- Oxidation and reduction always occur together.

Example :

$$Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$$

Oxidation :

$$Zn \rightarrow Zn^{2+} + 2e^-$$

(Zinc loses electrons → oxidised)

Reduction :

$$Cu^{2+} + 2e^- \longrightarrow Cu$$

(Copper(II) ion gains electrons → reduced)

Zn is oxidised & Cu^{2+} is reduced.

6. SUMMARY TABLE

Basis	Oxidation	Reduction
Meaning	Addition of oxygen / Removal of hydrogen / Loss of electrons	Removal of oxygen / Addition of hydrogen / Gain of electrons
Mnemonics	OIL (Oxidation Is Loss)	RIG (Reduction Is Gain)
Effect on oxidation number	Increases	Decreases
Example	$2Mg + O_2 \rightarrow 2MgO$	$CuO + H_2 \rightarrow Cu + H_2O$

7. IMPORTANT POINTS

- ★ Oxidation and reduction are opposite processes but occur together.
- ★ In nature, many reactions are redox, such as rusting of iron.
- ★ Redox reactions are very important in many industrial and biological processes.
- ★ Energy changes (heat, light, electricity) are often associated with redox reactions.

★ UNDERSTAND OXIDATION AND REDUCTION WELL – IT IS THE HEART OF MANY CHEMICAL REACTIONS! ★

CLASS 10 SCIENCE
CHEMICAL REACTIONS AND EQUATIONS

PART – 7

CORROSION AND RANCIDITY

When metals are attacked by substances in their environment slowly, they get destroyed. This process is called **corrosion**.
Fats and oils in food get oxidised by air and become harmful. This is called **rancidity**.

1. CORROSION

Corrosion is the slow destruction of metals by various substances such as air, moisture, acids, salts, etc.

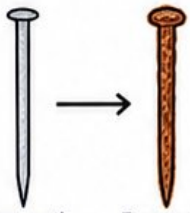
Example : Rusting of iron

Reaction : $4Fe + 3O_2 + xH_2O \longrightarrow 2Fe_2O_3 \cdot xH_2O$

(Reddish brown rust)

★ **Conditions necessary for rusting of iron :**

→ Presence of oxygen (air)
→ Presence of water (moisture)
→ Presence of an electrolyte (such as salt)



Iron nail (Before rusting) — Iron nail (After rusting)

EFFECTS OF CORROSION

★ Wastage of metals.
★ Loss of strength and life of metal articles.
★ Damage to machines, buildings, vehicles, bridges, etc.
★ Huge economic loss.



2. PREVENTION OF CORROSION

Corrosion can be prevented by the following methods :

① **Painting** : A coat of paint prevents air and moisture from coming in contact with the metal.



② **Oiling and Greasing** : A thin layer of oil or grease keeps air and moisture away.



③ **Galvanisation** : Coating a metal (iron) with a layer of zinc protects it from rusting.

$Fe + Zn \xrightarrow{\text{Galvanisation}} Zn\text{-coated Fe}$

Example : Iron sheets, nails, pipes.



④ **Alloying** : Making alloys of metals which are resistant to rusting.

Example : Stainless steel (iron + chromium)



⑤ **Electroplating** : Coating a metal with another metal (e.g., chromium, nickel, tin).



3. RANCIDITY

Rancidity is the oxidation of fats and oils in food, which causes a bad smell and taste and makes the food unfit for consumption.

Example : Butter, ghee, cooking oil, chips, dalda, etc.

Process : Fats and oils react slowly with oxygen from air and form peroxides, aldehydes, ketones and other compounds responsible for rancid smell.



Factors affecting rancidity :

→ Exposure to air (oxygen)
→ High temperature
→ Light
→ Presence of moisture
→ Presence of certain metal ions (e.g., Cu^{2+}, Fe^{2+})

4. PREVENTION OF RANCIDITY

Rancidity can be prevented by the following methods :

Antioxidants act by removing oxygen.

① **By using antioxidants** : Antioxidants like BHA, BHT, Vitamin C and Vitamin E slow down the process of oxidation.

② **By storing food in airtight containers** : This prevents the contact of food with air.



③ **By refrigeration** : Low temperature slows down the oxidation of fats and oils.



④ **By flushing with nitrogen gas** : Packets of chips, namkeen, etc., are flushed with nitrogen gas to prevent oxidation.



⑤ By using good quality oils and fats.

⑥ By keeping food away from light and moisture.

SUMMARY

Corrosion	Rancidity
★ Slow destruction of metals.	★ Oxidation of fats and oils.
★ Caused by air, moisture and electrolytes.	★ Causes bad smell and taste.
★ Can be prevented.	★ Can be prevented.

IMPORTANT POINTS

★ Corrosion is a redox reaction.
★ Rusting of iron is an example of corrosion.
★ Rancidity is also a type of oxidation.
★ Prevention of corrosion and rancidity saves money and keeps things useful for longer time.



PREVENTION IS BETTER THAN CURE – SAVE METALS, SAVE FOOD, SAVE FUTURE !

CLASS 10 SCIENCE
CHEMICAL REACTIONS AND EQUATIONS

PART - 8

IMPORTANT CONCEPTS, DIAGRAMS AND KEY POINTS

1. PHYSICAL STATES IN EQUATIONS

(s) Solid – e.g., $CaCO_3$ (s)

(l) Liquid – e.g., H_2O (l)

(g) Gas – e.g., CO_2 (g)

(aq) Aqueous solution – e.g., NaCl (aq)

Writing physical states is VERY IMPORTANT

2. BALANCING EQUATIONS – QUICK STEPS

1. Write the unbalanced (skeleton) equation.
2. Count the number of atoms of each element on both sides.
3. Start balancing one element at a time.
4. First balance metals, then non-metals.
5. Then balance hydrogen atoms.
6. Finally balance oxygen atoms.
7. Check again. Both sides must be equal.
8. Do not change the formula, only use coefficients.

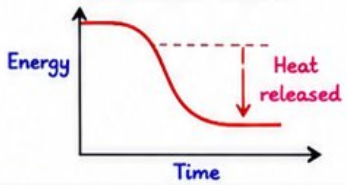
3. TYPES OF REACTIONS – QUICK RECAP

- ★ Combination — A + B → AB
- ★ Decomposition — AB → A + B
- ★ Displacement — A + BC → AC + B
- ★ Double Displacement — AB + CD → AD + CB
- ★ Precipitation (a type of double displacement)

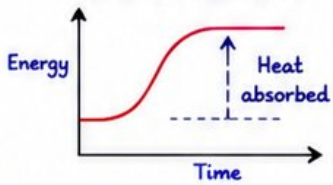
4. EXOTHERMIC vs ENDOTHERMIC REACTIONS

BASIS	EXOTHERMIC	ENDOTHERMIC
Heat energy	Released	Absorbed
Effect on surroundings	Surroundings feel hot	Surroundings feel cold
Temperature	Increases	Decreases
Nature	Generally fast	Generally slow
Examples	Combustion, neutralisation $CaO + H_2O$	Thermal decomposition, photosynthesis

EXOTHERMIC GRAPH



ENDOTHERMIC GRAPH



5. OXIDATION, REDUCTION & REDOX

- ★ Oxidation : Addition of oxygen / Removal of hydrogen / Loss of electrons.
- ★ Reduction : Removal of oxygen / Addition of hydrogen / Gain of electrons.
- ★ Redox Reaction : Oxidation and reduction occur simultaneously.

OIL RIG

O – Oxidation	R – Reduction
I – Is	I – Is
L – Loss	G – Gain

EXAMPLE : $CuO + H_2 \rightarrow Cu + H_2O$

CuO is reduced; H_2 is oxidised

6. CORROSION AND RANCDITY – QUICK POINTS

★ Corrosion : Slow destruction of metals by air, moisture, acids, salts, etc. (Example – Rusting of iron)



★ Conditions for rusting of iron :
1. Presence of oxygen (air)
2. Presence of water (moisture)
3. Presence of an electrolyte (salt)

★ Rancidity : Oxidation of fats and oils in food which causes bad smell and taste.

★ Factors : Air (oxygen), high temperature, light, moisture, presence of metal ions.

★ Prevention : Antioxidants, airtight containers, refrigeration, flushing with nitrogen gas, keeping away from light.



7. VERY IMPORTANT POINTS

1. A balanced equation obeys the law of conservation of mass.
2. In a balanced equation, number of atoms of each element is equal on both sides.
3. Chemical equations show only formulas, not physical conditions.
4. Coefficients are used to balance equations. (Do not change subscripts)
5. Energy changes occur in every chemical reaction.
6. Oxidation and reduction always occur together.
7. Corrosion and rancidity are examples of slow oxidation.

★ PRACTICE MORE, WRITE NEATLY, BALANCE CAREFULLY – SUCCESS IS SURE ! ★

CLASS 10 SCIENCE
CHEMICAL REACTIONS AND EQUATIONS

PART – 9

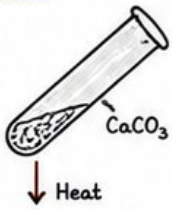
CASE-BASED QUESTIONS AND ASSERTION-REASON QUESTIONS

A. CASE-BASED QUESTIONS

Read the passage carefully and answer the questions that follow :

CASE – 1

Rohan heated calcium carbonate strongly in a test tube as shown in the given diagram. A colourless gas was evolved and a white residue was left behind.



Questions :

1. What type of reaction is taking place ?
2. Write the balanced chemical equation for the reaction.
3. What is the colourless gas evolved ?
4. What is the white residue left behind ?
5. Is this reaction endothermic or exothermic ?

Answers :

1. Decomposition reaction
2. $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
3. Carbon dioxide (CO_2)
4. Calcium oxide (CaO)
5. Endothermic reaction (heat is absorbed)

CASE – 2

Iron nails are kept in moist air for some days. A brown coating is observed on the surface of the nails. This coating can be removed by rubbing.



Questions :

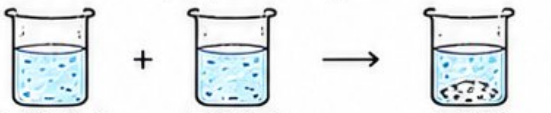
1. What is the brown coating formed on the iron nail called ?
2. Name the process by which it is formed.
3. Name two conditions necessary for this process.
4. Write the balanced equation for rusting of iron.
5. How can rusting of iron be prevented ?

Answers :

1. Rust
2. Corrosion (Rusting)
3. Presence of oxygen (air) and moisture (water)
4. $4Fe + 3O_2 + xH_2O \longrightarrow 2Fe_2O_3 \cdot xH_2O$ (Rust)
5. By painting, oiling/greasing, galvanisation, electroplating, alloying, etc.

CASE – 3

A student mixed silver nitrate solution with sodium chloride solution. A white precipitate was formed.



Questions :

1. What type of reaction is this ?
2. Write the balanced chemical equation.
3. What is the white precipitate formed ?
4. State one use of this precipitate.
5. Is this reaction exothermic or endothermic ?

Answers :

1. Precipitation reaction
2. $AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s)\downarrow + NaNO_3(aq)$
3. Silver chloride ($AgCl$)
4. Used in photography.
5. Neither exothermic nor endothermic (No heat change is noticeable)

B. ASSERTION-REASON QUESTIONS

In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option :

Options :

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

① **Assertion (A)** : Decomposition of water into hydrogen and oxygen is an endothermic reaction.
Reason (R) : Heat energy is absorbed in this reaction.
Answer : _____

② **Assertion (A)** : Rusting of iron is a redox reaction.
Reason (R) : Both oxidation and reduction occur during rusting.
Answer : _____

③ **Assertion (A)** : We should store food in airtight containers.
Reason (R) : This prevents rancidity of food.
Answer : _____

④ **Assertion (A)** : In the reaction $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$, zinc displaces copper.
Reason (R) : Zinc is more reactive than copper.
Answer : _____

KEY POINTS TO REMEMBER

- ★ Chemical reactions bring about changes in substances.
- ★ Balanced equations obey the law of conservation of mass.
- ★ Types : Combination, Decomposition, Displacement, Double displacement, Precipitation.
- ★ Energy changes : Exothermic (heat released), Endothermic (heat absorbed).
- ★ Oxidation : Add oxygen / Lose hydrogen / Lose electrons.
- ★ Reduction : Remove oxygen / Gain hydrogen / Gain electrons.
- ★ Redox reactions have both oxidation and reduction.
- ★ Corrosion and rancidity are slow oxidation processes.

PRACTICE CASE-BASED AND ASSERTION-REASON QUESTIONS REGULARLY TO SCORE HIGH IN EXAMS !

CLASS 10 SCIENCE
CHEMICAL REACTIONS AND EQUATIONS
PART – 10
MOST IMPORTANT PYQs AND HOTS QUESTIONS

A. PREVIOUS YEAR QUESTIONS (PYQs)

1. What is a balanced chemical equation ? Why should chemical equations be balanced ? (2019)

Ans. A balanced chemical equation has equal number of atoms of each element on both sides of the equation. Chemical equations should be balanced to obey the law of conservation of mass.

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2. State one observation when calcium carbonate is heated in a test tube. Write the balanced equation for the reaction. (2020)

Ans. Observation : A colourless gas is evolved which turns lime water milky.

Balanced equation :

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

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3. What is corrosion ? How can it be prevented ? (2021)

Ans. Corrosion is the slow destruction of metals by air, moisture, acids, etc.

Prevention : By painting, oiling/greasing, galvanisation, electroplating, using alloys, etc.

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4. Write the balanced equation for the reaction between zinc and dilute sulphuric acid. What type of reaction is it ? (2022)

Ans. $Zn + H_2SO_4(dil.) \longrightarrow ZnSO_4 + H_2\uparrow$

It is a single displacement reaction.

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5. Define oxidation and reduction with one example each. (2023)

Ans. Oxidation : Addition of oxygen / removal of hydrogen / loss of electrons.

Example : $2Mg + O_2 \longrightarrow 2MgO$

Reduction : Removal of oxygen / addition of hydrogen / gain of electrons.

Example : $CuO + H_2 \longrightarrow Cu + H_2O$

B. HOTS (HIGHER ORDER THINKING SKILLS) QUESTIONS

1. Why is respiration considered an exothermic reaction while photosynthesis is an endothermic reaction ?

Ans. In respiration, glucose reacts with oxygen and energy is released in the form of heat, hence it is exothermic. In photosynthesis, plants absorb sunlight energy to convert CO_2 and water into glucose, hence it is endothermic.

Respiration (Exothermic) → Heat released

Photosynthesis (Endothermic) → Heat absorbed

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2. Iron is more reactive than copper but less reactive than zinc. Explain with the help of an activity.

Ans. When an iron nail is placed in copper sulphate solution, no reaction occurs. But when placed in zinc sulphate solution, zinc displaces iron.

Thus, reactivity order : Zn > Fe > Cu

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3. Why do we store food items containing fats and oils in the refrigerator ?

Ans. Fats and oils get oxidised in air and become rancid. Low temperature in refrigerator slows down the oxidation process and prevents rancidity.

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4. Give reasons for the following :

(a) Balanced equations have the same number of atoms on both sides.

(b) Coefficients are changed, not subscripts, while balancing a chemical equation.

Ans.

(a) To obey the law of conservation of mass.

(b) Changing subscripts changes the chemical formula and identity of the substance.

C. FIVE MARKS PRACTICE QUESTIONS

1. Explain the reaction between lead nitrate solution and potassium iodide solution with a balanced equation. Write the type of reaction.

Ans. A yellow precipitate of lead iodide is formed.

$$Pb(NO_3)_2(aq) + 2KI(aq) \longrightarrow PbI_2(s)\downarrow + 2KNO_3(aq)$$

Type : Double displacement (Precipitation) reaction.

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2. Explain rusting of iron in detail and write the balanced equation for rusting.

Ans. Rusting occurs when iron reacts slowly with oxygen and moisture in the presence of an electrolyte (salt).

Balanced equation :

$$4Fe + 3O_2 + xH_2O \longrightarrow 2Fe_2O_3 \cdot xH_2O \text{ (Rust)}$$

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3. State and explain the difference between exothermic and endothermic reactions with examples.

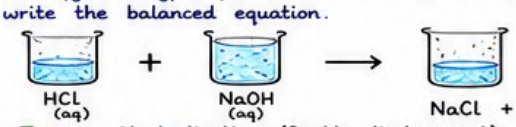
Ans. Refer to Part – 5 (Differences Table).

Examples : Exothermic – Combustion of methane

Endothermic – Thermal decomposition of $CaCO_3$

D. DIAGRAM-BASED QUESTIONS

1. Identify the type of reaction shown below and write the balanced equation.



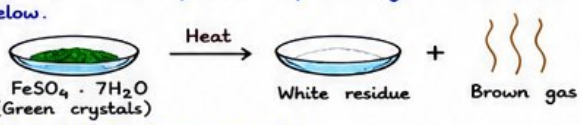
Ans. Type : Neutralisation (Double displacement)

Balanced equation :

$$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

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2. A student heats ferrous sulphate crystals as shown below.



(a) Name the brown gas evolved.

(b) Write the balanced equation for the reaction.

Ans. (a) Sulphur dioxide (SO_2)

(b) $2FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_2\uparrow + SO_3\uparrow$

E. FILL IN THE BLANKS

1. The addition of oxygen is called _____.
2. The loss of electrons is called _____.
3. The process of coating iron with zinc is called _____.
4. Food gets rancid due to _____ of fats and oils.
5. CO_2 turns lime water _____.

F. TRUE / FALSE

1. Rusting of iron is a fast process. ( )
2. Balanced equations obey law of conservation of mass. ( )
3. Exothermic reactions absorb heat. ( )
4. Respiration is an endothermic reaction. ( )
5. Photosynthesis is an endothermic reaction. ( )

G. MATCH THE FOLLOWING

Column A	Column B
1. $CaCO_3$ (heated)	(a) Exothermic
2. Burning of LPG	(b) Endothermic
3. Rusting of iron	(c) Corrosion
4. Photosynthesis	(d) Endothermic

Ans. 1-b, 2-a, 3-c, 4-d

REVISION TODAY, SUCCESS TOMORROW !

CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

PART – 11

FULL CHAPTER REVISION – MIND MAP

1. CHEMICAL EQUATION

- Representation of a chemical reaction using symbols and formulae.
- Shows reactants ⟶ products
- Must be balanced.

Types :

- Word equation – in words
- Skeletal equation – formulas without balancing
- Balanced equation – equal no. of atoms on both sides

Example (Balanced) :

$2H_2 + O_2 \longrightarrow 2H_2O$

2. BALANCING EQUATIONS – STEPS

1. Write the unbalanced equation.
2. Count atoms of each element on both sides.
3. Start balancing one element at a time.
4. First balance metals, then non-metals.
5. Then balance H atoms.
6. Finally balance O atoms.
7. Check again. Both sides must be equal.
8. Do not change formula, only use coefficients.

3. TYPES OF CHEMICAL REACTIONS

★ **Combination**
$A + B \longrightarrow AB$

★ **Decomposition**
$AB \longrightarrow A + B$

★ **Displacement**
$A + BC \longrightarrow AC + B$

★ **Double displacement**
$AB + CD \longrightarrow AD + CB$

★ **Precipitation** (type of double displacement)

4. EXOTHERMIC & ENDOTHERMIC REACTIONS

(ii) Exothermic

- Heat is released
- Surroundings feel hot
- Temperature increases
- Generally fast

Example : Combustion, neutralisation

$CaO + H_2O \longrightarrow Ca(OH)_2 + \text{Heat}$

(ii) Endothermic

- Heat is absorbed
- Surroundings feel cold
- Temperature decreases
- Generally slow

Example : Thermal decomposition, Photosynthesis

CHEMICAL REACTIONS AND EQUATIONS

5. OXIDATION, REDUCTION AND REDOX

- **Oxidation :** Add O / Lose H / Lose e^-
- **Reduction :** Remove O / Gain H / Gain e^-
- **Redox reaction :** Oxidation and reduction occur simultaneously.

OIL RIG

O – Oxidation	R – Reduction
I – Is	I – Is
L – Loss	G – Gain

Example : $CuO + H_2 \longrightarrow Cu + H_2O$

- CuO is reduced
- H_2 is oxidised

6. CORROSION & RANCIDITY

Corrosion

- Slow destruction of metals by air, moisture, acids, salts.
- Example : Rusting of iron
- Conditions for rusting :
  1. Presence of oxygen (air)
  2. Presence of moisture (water)
  3. Presence of an electrolyte (salt)
- Prevention : Painting, oiling/greasing, galvanisation, electroplating, using alloys.

Rancidity

- Oxidation of fats and oils in food.
- Causes bad smell and taste.
- Factors : Air (oxygen), high temperature, light, moisture, metal ions.
- Prevention : Antioxidants, airtight containers, refrigeration, nitrogen flushing, away from light.

7. PHYSICAL STATES IN EQUATIONS

(s)	Solid	e.g., $CaCO_3$ (s)
(l)	Liquid	e.g., H_2O (l)
(g)	Gas	e.g., CO_2 (g)
(aq)	Aqueous solution	e.g., NaCl (aq)

Writing physical states is VERY IMPORTANT

8. IMPORTANT CONCEPTS

★ Chemical reactions produce new substances.
★ Atoms are neither created nor destroyed.
★ Balanced equations obey law of conservation of mass.
★ Energy changes occur in every reaction.
★ Oxidation and reduction always occur together.
★ Corrosion and rancidity are slow oxidation processes.
★ Coefficients are used to balance equations.
★ Chemical reactions are useful in daily life.

9. KEY WORDS

→ Reactants – substances which react
→ Products – substances formed
→ Coefficient – number written before formula
→ Oxidation – addition of O / loss of H / loss of e^-
→ Reduction – removal of O / gain of H / gain of e^-
→ Exothermic – heat released
→ Endothermic – heat absorbed
→ Corrosion – rusting of metals
→ Rancidity – spoilage of food due to oxidation of fats and oils

10. QUICK FORMULAS

- No. of atoms of an element = (Subscript) × (Coefficient)
- Balanced equation : No. of atoms (LHS) = No. of atoms (RHS)
- General forms :

Combination	$A + B \longrightarrow AB$
Decomposition	$AB \longrightarrow A + B$
Displacement	$A + BC \longrightarrow AC + B$
Double displacement	$AB + CD \longrightarrow AD + CB$

11. EXAMPLES TO REMEMBER

(i) Combination : $2Mg + O_2 \longrightarrow 2MgO$
(ii) Decomposition : $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
(iii) Displacement : $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$
(iv) Double displacement : $AgNO_3 + NaCl \longrightarrow AgCl\downarrow + NaNO_3$
(v) Precipitation : $BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4\downarrow + 2NaCl$

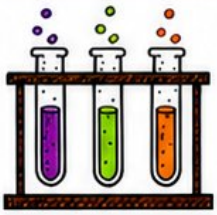
12. SUMMARY (ONE LINE EACH)

★ Equations show what happens in a reaction.
★ Balanced equations follow law of conservation of mass.
★ Types of reactions help in finding products.
★ Exothermic releases heat, endothermic absorbs.
★ Oxidation and reduction occur together (redox).
★ Corrosion and rancidity damage useful things.
★ Prevention is better than cure!

UNDERSTAND – PRACTICE – REVISE = SUCCESS IN EXAMS



CLASS 10 SCIENCE
CHEMICAL REACTIONS AND EQUATIONS



PART - 12

★ ULTRA QUICK REVISION SHEET - 1 PAGE BEFORE EXAM ★

1. CHEMICAL EQUATION

- Representation of a chemical reaction using symbols and formulae.
- Shows reactants ⟶ products
- Must be balanced.

Example (Balanced) :

$$2H_2 + O_2 \longrightarrow 2H_2O$$

2. BALANCING EQUATION - QUICK STEPS

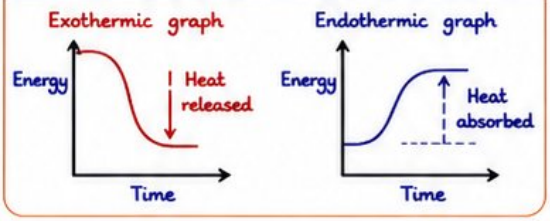
① Write the unbalanced equation.
② Count atoms of each element on both sides.
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④ First balance metals, then non-metals.
⑤ Then balance H atoms.
⑥ Finally balance O atoms.
⑦ Check again. Both sides must be equal.
⑧ Do not change formula, only coefficients.

3. TYPES OF REACTIONS

★ Combination : $A + B \longrightarrow AB$
★ Decomposition : $AB \longrightarrow A + B$
★ Displacement : $A + BC \longrightarrow AC + B$
★ Double displacement : $AB + CD \longrightarrow AD + CB$
★ Precipitation : (type of double displacement) Insoluble salt (precipitate) is formed.

4. EXOTHERMIC vs ENDOTHERMIC

Basis	Exothermic	Endothermic
Heat energy	Released	Absorbed
Effect on surroundings	Feel hot	Feel cold
Temperature	Increases	Decreases
Nature	Generally fast	Generally slow
Examples	Combustion, Neutralisation ($CaO + H_2O$)	Thermal decomposition, Photosynthesis



5. OXIDATION, REDUCTION & REDOX

- Oxidation : Add O / Lose H / Lose e^-
- Reduction : Remove O / Gain H / Gain e^-
- Redox reaction : Oxidation and reduction occur simultaneously.

OIL RIG

O - Oxidation	R - Reduction
I - Is	I - Is
L - Loss	G - Gain

Example : $CuO + H_2 \rightarrow Cu + H_2O$
(CuO is reduced; H_2 is oxidised)

6. CORROSION & RANCIDITY

Corrosion

- Slow destruction of metals by air, moisture, acids, salts.

Conditions for rusting of iron :
① Presence of O_2 (air)
② Presence of H_2O
③ Presence of electrolyte (salt)

Prevention : Painting, oiling/greasing, galvanisation, electroplating, using alloys.

Rancidity

- Oxidation of fats and oils in food.

Factors :
- Air (O_2)
- High temperature
- Light
- Moisture
- Metal ions (e.g. Cu^{2+}, Fe^{2+})

Prevention : Antioxidants, airtight containers, refrigeration, N_2 flushing, away from light.

7. PHYSICAL STATES IN EQUATIONS

(s) Solid e.g., $CaCO_3$ (s)
(l) Liquid e.g., H_2O (l)
(g) Gas e.g., CO_2 (g)
(aq) Aqueous solution e.g., NaCl (aq)

Writing physical states is VERY IMPORTANT

8. KEY WORDS

→ Reactants - substances which react
→ Products - substances formed
→ Coefficient - number written before formula
→ Oxidation - add O / lose H / lose e^-
→ Reduction - remove O / gain H / gain e^-
→ Exothermic - heat released
→ Endothermic - heat absorbed
→ Corrosion - rusting of metals
→ Rancidity - spoilage of food due to oxidation of fats and oils

9. IMPORTANT FORMULAS / EQUATIONS

① $2H_2 + O_2 \longrightarrow 2H_2O$
② $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
③ $Zn + H_2SO_4$ (dil.) $\longrightarrow ZnSO_4 + H_2\uparrow$
④ $4Fe + 3O_2 + xH_2O \longrightarrow 2Fe_2O_3 \cdot xH_2O$ (Rust)
⑤ $CuO + H_2 \longrightarrow Cu + H_2O$
⑥ $BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4\downarrow + 2NaCl$

10. DIFFERENCE BETWEEN EXO & ENDO (ONE LINE)

Exothermic releases heat to surroundings. Temperature increases.
VS
Endothermic absorbs heat from surroundings. Temperature decreases.

11. VERY IMPORTANT POINTS

★ Balanced equations obey the law of conservation of mass.
★ Chemical reactions show only formulas, not physical conditions.
★ Types of reactions help in finding products.
★ Oxidation and reduction always occur together.
★ Prevention is better than cure !

12. QUICK CHECK - 5 SECCONDS

Equation representation | Balance atoms | Types (5) | Exo- heat out, Endo- heat in | Redox- OIL RIG, Oxidation / Reduction | Corrosion-rusting, Rancidity-spoilage | Physical states (s) (l) (g) (aq) | Coefficients not subscripts

★ STAY CALM ★ REVISE SMART ★ ATTEMPT WELL ★ SCORE 100/100 ★